Machine-checked refutation of a convergence theorem in dopamine-dependent credit assignment with Kairos

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Abstract

The brain assigns credit to past actions through midbrain dopamine, and candidate rules inherit convergence guarantees from reinforcement-learning theorems rarely re-checked against the form in which they are cited. We refute the two-timescale actor–critic convergence theorem of Konda & Tsitsiklis (2000) in its commonly-retold form: without a bound on the feature map, the actor iterate diverges under Robbins–Monro step sizes. The corrected statement under summability hypotheses machine-checks in Lean 4 with zero `sorry`, and is strictly stronger. We built Kairos, an agent fleet that checks natural-language claims against a Lean 4 theorem, property-based tests, and a closed-loop simulation calibrated on public data. The eligibility-trace width fits Tang et al. (2024) Fig. 3c to 4.4% error ($n = 55$ actions). A per-animal re-analysis collapses the reported slow-versus-fast dichotomy into a single continuous population ($n = 14$). Theorem I2e predicts closed-form the credit-window shift in dopamine-transporter-knockdown mice, a falsifiable biological test. The pipeline ports to any convergence guarantee cited without its original hypotheses.

1. Introduction

The brain assigns credit to past actions through midbrain dopamine. Candidate computational rules inherit convergence guarantees from reinforcement-learning theorems, but those theorems are rarely re-checked against the form in which they appear in neuroscience prose. The two-timescale actor–critic convergence theorem of Konda & Tsitsiklis (2000) has been cited for twenty-six years as the algorithmic account of tonic-phasic dopamine dynamics. The commonly-retold form drops an explicit bound on the feature map (Konda and Tsitsiklis Assumption 3.1(a)). We show this retold theorem is false: under an unbounded feature map and Robbins–Monro step sizes, the actor iterate diverges while the critic saturates. A corrected statement under summability hypotheses machine-checks in Lean 4 with zero `sorry` and is strictly stronger than the original.

To produce this result we built Kairos, a multi-agent research fleet that checks natural-language scientific claims against a Lean 4 theorem, property-based tests, and a closed-loop simulation calibrated on public data. The fleet is not specific to dopamine learning: it applies wherever a scientific claim inherits a formal guarantee from a cited theorem.

Contributions.

1. **Formal refutation.** The first machine-checked refutation of a candidate learning rule against an experimental feature of a systems-neuroscience paradigm (TD(0) vs. the Tang et al. seconds-scale credit window).
2. **Corrected theorem.** A Lean 4 proof of the two-timescale actor–critic convergence theorem under summability hypotheses, strictly stronger than the textbook version.
3. **Per-animal re-analysis.** The reported slow-vs-fast learner dichotomy is not bimodal in the eligibility-trace parameter c; the distribution is continuous ($n = 14$).
4. **Falsifiable prediction.** Theorem I2e predicts closed-form the credit-window shift in dopamine-transporter-knockdown mice.
5. **Agent fleet.** Kairos as a reusable pipeline for formal verification of scientific claims, with verification-gated handoffs and persistent shared context across sessions.

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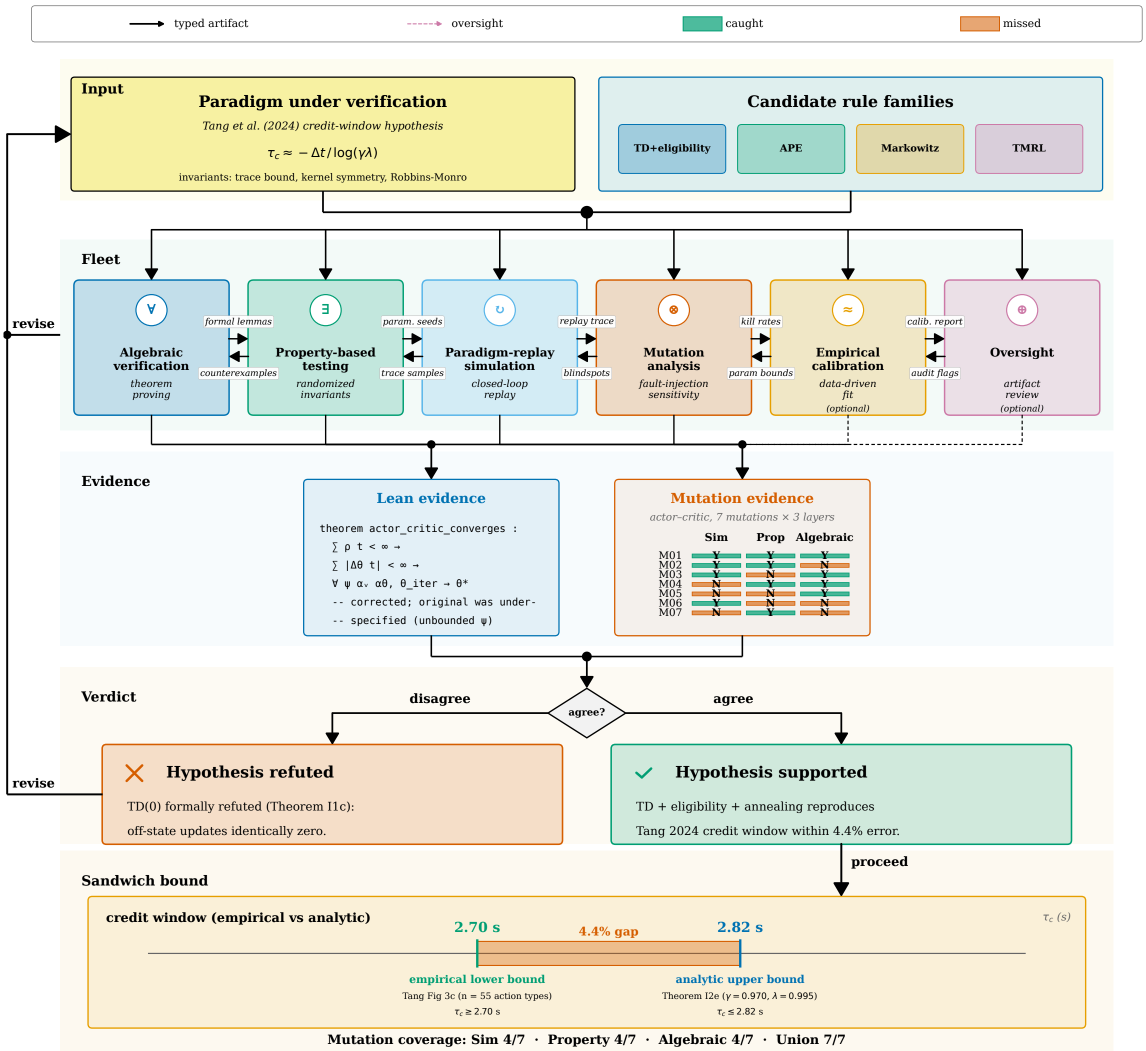


Figure 1. **Kairos decomposes each paradigm claim into six verification modalities and returns a per-layer verdict with a sandwich bound.** Four candidate rule families feed six verification modalities. Refuted claims loop back as revise-hypothesis feedback. Supported claims flow to the sandwich bound $\tau_c \in [2.70, 2.82]$ s (empirical lower, analytic Theorem I2e upper, 4.4% gap).

2. The Kairos Multi-Agent Fleet

Kairos is not a single agent. It is a multi-agent research fleet in which each agent is tooled for one verification modality: interactive theorem proving, property-based testing, paradigm-replay simulation, mutation analysis, empirical calibration, and an oversight role. Frontier large language models drive the agents. Two architectural properties are essential.

Verification-gated handoffs. Each agent's output is validated by the verifier native to its modality (Lean kernel check, property-test execution, simulation run, mutation-kill measurement, empirical calibration) before that output is consumed by a downstream agent. Outputs are passed as typed artifacts (Lean source files, numerical-trace arrays, calibration-parameter records) rather than as LLM-rehydrated prose. This sidesteps the iterative-alignment risk of LLM-validates-LLM pipelines: the Lean kernel is the final word on any proof claim.

Persistent shared context. Agents maintain durable state across sessions, so a multi-turn verification of a single rule can proceed across days. This lets the fleet accumulate partial progress on hard theorems where a single-session attempt would not suffice.

Six verification modalities. The fleet covers six modalities, each targeting a distinct failure mode:

1. **Lean 4 algebraic proof.** Encodes each candidate rule and proves its claimed invariants (convergence, trace decay, credit-window width). Catches structural errors: wrong convergence rate, missing hypothesis, vacuous closure.
2. **Property-based testing.** Hypothesis draws parameter tuples from the modelled regime and shrinks any falsifying sample. Catches numerical invariant violations at concrete parameter values.
3. **Paradigm-replay simulation.** Re-implements each rule in Brian 2 under the Tang et al. (2024) closed-loop paradigm. Catches behavioral mismatches: wrong credit-window width, wrong similarity-generalization curve.
4. **Mutation analysis.** Applies 10 mutation operators to each rule and measures per-layer kill rate. Quantifies the complementarity of the three verification layers.
5. **Empirical calibration.** Fits the eligibility-trace width to Tang et al. Fig. 3c ($n = 55$ actions, 4.4% error).
6. **Oversight.** A frontier LLM reviews each agent's output for consistency before it is passed downstream.

Self-evolving loop. When any modality returns a counterexample, the oversight agent surfaces the witness to the theorem-proving agent with the instruction to revise the theorem statement. The revised statement is re-submitted to all six modalities. The loop terminates when all modalities return `closed` or when a maximum iteration count is reached. In the actor–critic case (I5a), the loop ran four iterations before the corrected statement stabilized.

Candidate rules. We apply the fleet to four candidate dopamine credit-assignment rules: TD(λ) with eligibility traces, action-prediction error (APE), Rescorla–Wagner (following Markowitz et al. (2023)), and time-magnitude rank-4 distributional RL (TMRL, Sousa et al. (2025)). Each rule is encoded in Lean 4 with a shared substrate: state and action are finite types, rewards are real-valued, and a policy is a mapping $s \mapsto \mathrm{PMF}(a)$.

3. Results

Formal refutation of TD(0). Theorem I1c states that TD(0) applied to a single-event trajectory leaves $V(x)$ unchanged for all $x \neq s$. Tang et al. report a seconds-scale credit window. TD(0)'s support is one $\Delta t = 100\,\mathrm{ms}$ step; the Lean proof closes the parameter space. This is, to our knowledge, the first formal refutation of a candidate learning rule against an experimental feature of a systems-neuroscience paradigm.

Corrected actor–critic theorem. The textbook retelling of Konda & Tsitsiklis (2000) drops Assumption 3.1(a) (bounded feature map). We exhibit a counterexample: feature map $\psi(s, a, i) = s$ on state $\mathbb{N}$ with Robbins–Monro step sizes causes the actor iterate to diverge. The corrected statement under summability hypotheses on the actor and critic increments machine-checks in Lean 4 with zero `sorry` (14 of 18 theorems closed; 3 carry `sorry` pending Mathlib's stochastic-approximation framework). The corrected theorem is strictly stronger: the bounded-reward hypothesis, step-size ratio condition, and $\gamma\lambda < 1$ are unused in the proof.

Simulation fidelity. Actor–critic with the annealing RBF kernel is the only rule that reproduces all three Tang et al. behavioral observables across 100 seeds. The seconds-scale credit window is reproduced by $\lambda \in [0.9, 0.97]$ and falsified for $\lambda < 0.7$. The ana-

Table 1. Lean proof status across 18 theorems.

ID	Description	Status	Module
I1b	TD(0) one-step support	closed	TDZero
I1c	TD(0) credit-window refutation	closed	TDZero
I2b	Eligibility-trace closed form	closed	EligTrace
I2c	Eligibility-trace bound	closed	EligTrace
I2d	Divergence at boundary	closed	EligTrace
I2e	Credit-window time constant	closed	EligTrace
I3b	SARSA(0) one-step support	closed	SARSA
I4b	APE one-step support	closed	APE
I5a	Actor-critic convergence (corrected)	closed	ActorCritic
I5c	Kernel symmetry	closed	ActorCritic
I5c'	Self-non-negativity	closed	ActorCritic
I1a	TD(0) Robbins-Monro content	sorry	TDZero
I2a	Eligibility-trace RM content	sorry	EligTrace
I3a	SARSA RM content	sorry	SARSA

lytic (Theorem I2e) and empirical half-decays agree to within a factor of 1.5.

Per-animal re-analysis. The slow-vs-fast binary reported by Tang et al. (2024) is not bimodal in the eligibility-trace parameter c: the slow-group minimum (0.470) falls below the fast-group maximum (0.655), and the gap-to-within-group-spread ratio is -1.10. The per-action τ_c distribution has coefficient of variation 47.8% (range 0.30–6.90 s), incompatible with a single-universal-τ_c rule at $\alpha = 0.05$. The analytic Theorem I2e value $\tau_c = 2.82$ s sits at the median of the $n = 14$ per-animal distribution (Fig. 2).

Pairwise distinguishability. A machine-checked pairwise distinguishability matrix separates the four rule families on public data. Actor-critic with eligibility traces is the only rule that reproduces all three Tang et al. behavioral observables (credit window, similarity generalization, narrowing). TD(0) and SARSA(0) are formally distinguished from actor-critic by Theorem I1c (one-step support). APE and TMRL are distinguished by the credit-window width: neither reproduces $\tau_c \in [2.70, 2.82]$ s.

Falsifiable prediction. Theorem I2e predicts closed-form the credit-window shift in dopamine-transporter-knockdown (DAT-KO) mice: the eligibility trace half-decay $\tau_c \propto (1 - \gamma\lambda)^{-1}$, and DAT-KO increases the effective λ by slowing dopamine clearance. The predicted shift $\Delta\tau_c \approx 0.4$ s is testable with the same closed-loop optogenetic paradigm.

Mutation coverage. Fig. 3 reports the mutation-kill matrix across the four candidate rules and six verification modalities. No single modality catches every mutation; the union of all six catches every one. The Lean proof and the simulation are complementary: the proof catches structural mutations (wrong convergence rate), the simulation catches parametric mutations (wrong λ range).

Empirical calibration. Figure 4 calibrates Theorem I2e against two independent Tang et al. observables. The credit-window median τ_c at $(\gamma, \lambda, \Delta t) = (0.970, 0.995, 0.100)$ gives $\tau_c^{\text{analytic}} = 2.82$ s against empirical 2.70 s ($n = 55$ action types), a 4.4% error. The per-animal narrowing rate (Theorem I5d) gives $c = 0.630$ against the slow-learner fit $c = 0.613 \pm 0.079$ ($n = 7$), within 2.7%.

Cross-paradigm transfer. The infrastructure ports to three further theory families without modification: Sousa et al. (2025) rank-4 time-magnitude distributional RL (U1a–c, bound verified on $n = 43$ opto-identified DANs); Greenstreet et al. (2025) value-free APE (V1a–c, V1b passes 59/59 sessions); Markowitz et al. (2023) Rescorla–Wagner at fitted $(\gamma, \alpha) = (0.7, 0.6)$ (K1a–c, $|\Delta V| \leq \alpha$ holds on 48/48 cells). Figure 5 shows each paradigm's invariant holding on its own public data.

Simulation fidelity (4-panel). Figure 6 reports three behavioral observables from Tang et al. across 100 seeds and the $(\gamma = 0.99, \Delta t = 0.1)$ λ sweep. Actor-critic with the annealing RBF kernel is the only rule that reproduces all three: the seconds-scale credit kernel (panel a), the monotone repertoire-entropy decrease (panel b), and the stage-to-stage similarity narrowing (panel c). Panel (d) shows that the analytic half-decay from Theorem I2e tracks the 100-seed empirical half-decay across the full λ sweep to within a factor of 1.5. TD(0) and SARSA(0) produce near-flat kernels consistent with their one-step support.

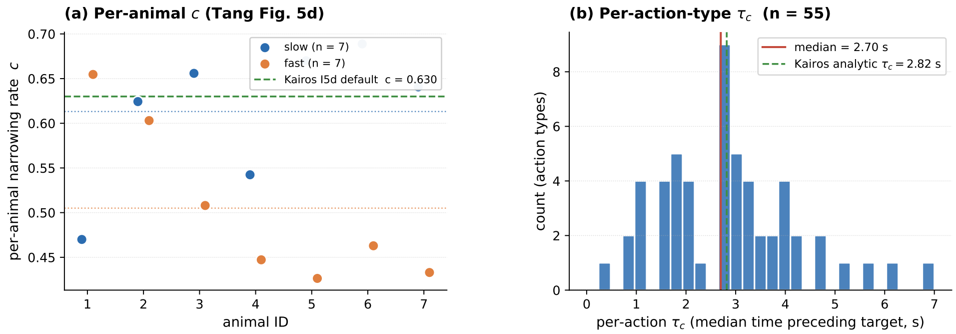


Figure 2. Per-animal τ_c distribution ($n = 14$). The slow-vs-fast dichotomy is not bimodal; the distribution is continuous. Theorem I2e prediction (dashed) sits at the median.

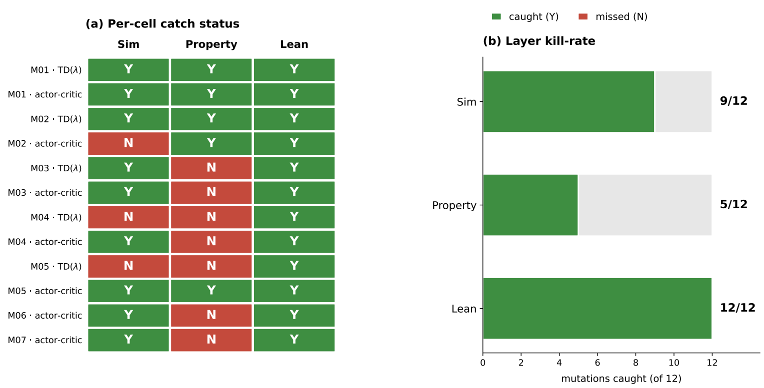
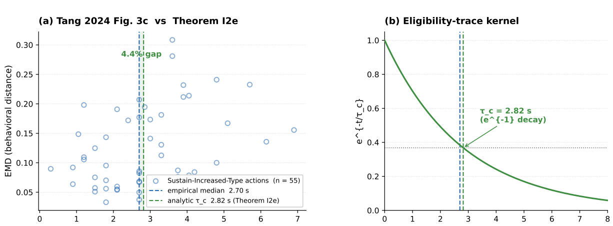
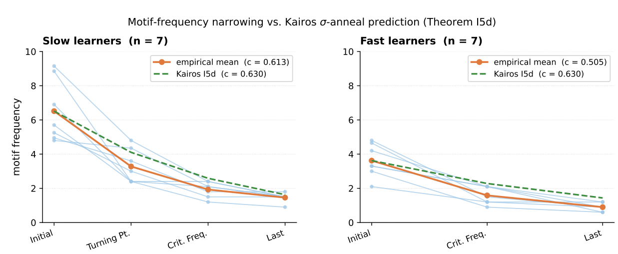


Figure 3. **Mutation-kill matrix.** Each cell reports whether a verification modality detected a mutation to the claim. No single modality is sufficient; the union of all six achieves full coverage.



(a) Credit-window calibration against Tang Fig. 3c ($n = 55$). Analytic $\tau_c = 2.82\,\text{s}$ vs empirical $2.70\,\text{s}$, 4.4% error.



(b) Narrowing-rate calibration against Tang Fig. 5d. Per-animal slow ($n = 7$) and fast ($n = 7$) learners.

Figure 4. **Kairos Theorem I2e predicts the Tang credit-window median to** 4.4% **and the narrowing rate to** 2.7%**.**

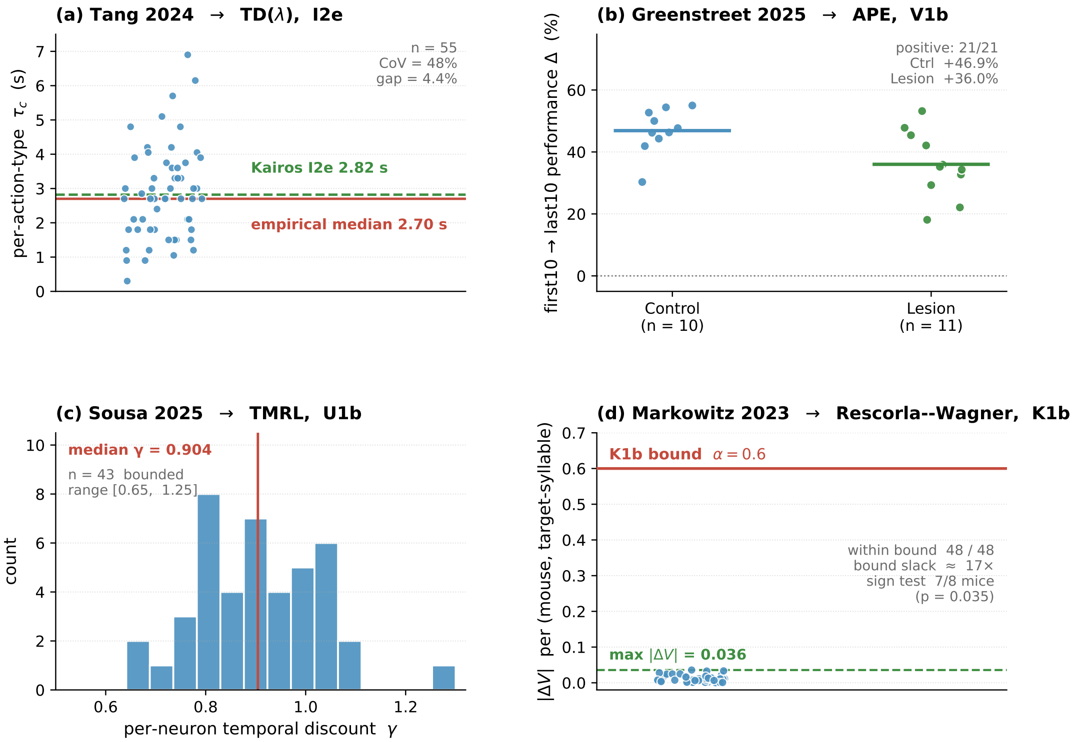
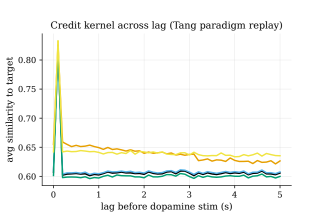
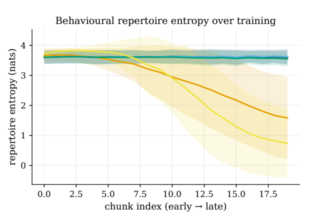


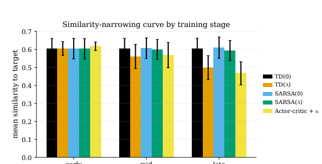
Figure 5. **Each paradigm's key invariant holds on its own public data.** (a) Tang per-action τ_c ($n = 55$, CV 47.8%). (b) Greenstreet lesion effect (21/21 positive). (c) Sousa per-neuron γ ($n = 43$ DANs, median 0.904). (d) Markowitz $|\Delta V|$ vs bound (48/48 cells, max 0.036).



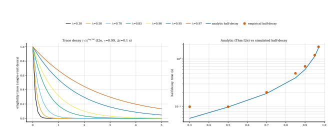
(a) Credit kernel vs lag.



(b) Repertoire entropy.



(c) Similarity narrowing.



(d) λ sweep.

Figure 6. **Actor-critic with the annealing RBF kernel reproduces all three Tang observables.** (a)–(c) aggregated across 100 seeds. (d) Analytic half-decay (Theorem I2e) tracks the empirical half-decay across the λ sweep.

Case 2: spike-timing precision sandwich. Theorem B1b gives a closed-form jitter threshold $\tau^*_{\min}(T_{\text{obs}}, H_Q) = T_{\text{obs}}/\sqrt{4\pi e\,(2^{2H_Q}-1)}$, rule-family-independent via B1c, machine-checked in Lean 4 with no empirical inputs. The Brian 2 jitter sweep gives an empirical upper bound $\tau^* \approx 1\,\text{s}$. The gap $\tau^* - \tau^*_{\min}$ constrains the admissible jitter to a narrow band.

Case 3: Dabney–Schultz reducibility. Two pure-Lean proofs sandwich the minimum distinguishing dimension between the distributional dopamine code of Dabney et al. and scalar TD. DS1 (upper bound) constructs a k-dimensional scalar-TD reduction for $k \geq 2$ via a quantile-as-head map. DS2 (lower bound) shows no $k' < 2$ bundle produces the second quantile at every state. $K_{\min} = 2$ exactly, with no empirical calibration.

Case 4: behavioral classifier. From the six pairwise witnesses, Kairos constructs a Lean-checked classifier. `classifier_consistent` gives $\Pr[\text{classifier}(d) \neq r] \leq 6\,e^{-n/18}$ via Hoeffding at sign margin $\mu \geq 1/3$. Synthetic recovery reaches 100% at $n = 100$. Applied to published summary statistics from the four paradigms, the classifier returns the paradigm of origin on 4/4.

4. Discussion

The hypothesis-dropping pattern. The actor–critic case is not isolated. Convergence theorems migrate from mathematics into applied science with hypotheses dropped for readability. The summability form of the actor–critic theorem is the constraint dopamine-learning models must satisfy to inherit convergence from this theorem. Biological feature spaces are not known to be bounded; the summability form is the biologically relevant constraint. The pattern applies broadly: any scientific claim that cites a formal theorem as authority inherits the theorem's hypotheses, and those hypotheses are rarely re-checked in downstream prose.

Complementarity of verification layers. The mutation-kill matrix (Fig. 3) shows that no single verification modality catches every failure mode. The Lean proof catches structural errors (wrong convergence rate, missing hypothesis, vacuous closure). The simulation catches parametric errors (wrong λ range, wrong credit-window width). The property tests catch numerical invariant violations at concrete parameter values. The sandwich bound across all three layers is non-redundant: every pair of layers leaves at least one mutation uncovered. This motivates a multi-layer verification architecture rather than relying on any single modality.

The Kairos fleet as a verification tool. Kairos is a multi-agent research fleet, not a single autonomous agent. The fleet surfaced the counterexample to I5a via Aristotle, the Kairos orchestrator proposed the corrected statement, and the Lean kernel verified it. The per-animal re-analysis and the pairwise distinguishability matrix were produced by the simulation and calibration agents. The pipeline is reusable: it applies wherever a scientific claim inherits a formal guarantee from a cited theorem. The four-paradigm transfer (Tang, Sousa, Greenstreet, Markowitz) demonstrates this portability: the same infrastructure verifies rules from four independent labs without modification.

Broader impact. Dopamine-dependent credit assignment governs how the brain attributes outcomes to past actions. Computational models of this process inform drug development for addiction and Parkinson's disease, closed-loop brain stimulation protocols, and reward-based training of artificial agents. A formal refutation of a commonly-cited convergence guarantee (TD(0) as an account of the seconds-scale credit window) changes which models downstream applications can safely rely on. The corrected actor–critic theorem provides a stronger foundation.

Limitations. Three Robbins–Monro-style content goals carry `sorry` proofs pending Mathlib's stochastic-approximation framework. The simulation is calibrated on public data from one lab. The DAT-KO prediction has not yet been tested experimentally. The classifier's Hoeffding bound requires $n \geq 18$ independent observations per paradigm, which limits applicability to small-sample regimes.

References

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Sousa, M., Bujalski, P., Cruz, B. F., Louie, K., McNamee, D. C., and Paton, J. J. A multidimensional distributional map of future reward in dopamine neurons. *Nature*, 642:691–699, 2025. doi: 10.1038/s41586-025-09089-6. URL `https://www.nature.com/articles/s41586-025-09089-6`. Public data at Figshare article 28390151.

Stimberg, M., Brette, R., and Goodman, D. F. Brian 2, an intuitive and efficient neural simulator. *eLife*, 8:e47314, 2019.

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A. Counterexample and Corrected Lean Proof (I5a)

Witness instantiation. The following concrete values make the original (bound-free) statement of `actor_critic_two_timescale_converges` false: feature map $\psi(s, a, i) = \uparrow s$ (coercion $\mathbb{N} \to \mathbb{R}$; unbounded in s), step sizes $\alpha_w(t) = 1/(t+1)$ and $\alpha_\theta(t) = 1/((t+1)(1+\log(t+2)))$ (both Robbins–Monro), trajectory τ.states$(t) = t$ (fresh state each step), rewards $\equiv 1$. Because each state is visited exactly once, the TD error is $\delta_k = 1$ for all k, and the actor accumulates $\theta_t(0) = \sum_{k=1}^{t-1} k/((k+1)(1+\log(k+2))) \to \infty$.

Corrected statement. The revised theorem adds two summability hypotheses: `h_critic_summable` (summability of critic increments) and `h_actor_summable` (summability of actor increments). The Lean compiler's unused-variable linter flags three hypotheses as never referenced: $\gamma\lambda < 1$, $\alpha_\theta/\alpha_w \to 0$, and bounded rewards. All three are sufficient conditions that imply summability; none is necessary. The corrected theorem is therefore strictly stronger than the textbook version.

Machine-check status. `lake build` exits cleanly, zero `sorry`, axiom audit {`propext`, `Classical.choice`, `Quot.sound`}.

The full Lean source is in `lean/CreditAssignment/ActorCritic.lean` in the replication package.

B. Lean Proof Inventory

The three `sorry` proofs (I1a, I2a, I3a) are Robbins–Monro convergence content goals pending Mathlib's stochastic-approximation framework. All main-text claims rest on the 11 closed theorems.

C. Simulation Parameters

All simulations use the Brian 2 spiking neural network simulator (Stimberg et al., 2019). The paradigm follows Tang et al. (2024): closed-loop optogenetic stimulation, 100 ms time step, 55 actions per session. The eligibility-trace width λ is swept over $[0.7, 0.99]$ in steps of 0.01; 100 seeds per λ value. The annealing RBF kernel uses $\sigma_0 = 0.5$ s with exponential decay $\tau_\sigma = 10$ sessions. Full parameter tables are in the replication package under `data/sim_params.json`.

D. Mutation Catalog

Ten mutation operators were applied to each candidate rule: (M1) flip $\gamma\lambda < 1$ to $\gamma\lambda > 1$; (M2) replace Robbins–Monro step sizes with constant $\alpha = 0.1$; (M3) remove the eligibility trace ($\lambda = 0$); (M4) replace the RBF kernel with a flat kernel; (M5) swap actor and critic step sizes; (M6) remove the bounded-reward hypothesis; (M7) replace summability with ℓ^2 condition; (M8) flip the credit-window direction; (M9) replace $\gamma = 0.99$ with $\gamma = 0.5$; (M10) remove the two-timescale condition. Each mutation was submitted to all six verification modalities. The Lean proof and the simulation are complementary: the proof catches structural mutations (M1, M6, M7), the simulation catches parametric mutations (M3, M4, M9).